Post-Quantum Cryptography Migration

Part 12 - Where It Runs: Security & Network Appliances

Compiled by the Spaced Research Ledger

Last updated: 2026-08-29

The boxes that guard networks, firewalls, load balancers, routers, and encryptors, have become one of the migration's busiest fronts. Every major vendor now ships or roadmaps post-quantum features, and some sell hardware designed around the new algorithms from the silicon up.

This report covers the appliance world in six sections. Section 1 covers the next-generation firewalls, Section 2 the load balancers, and Section 3 the routers, switches, and SD-WAN platforms. Section 4 covers the awkward physics of bigger handshakes, Section 5 Wi-Fi, and Section 6 the VPN concentrators and optical encryptors.

The competitive dynamic is real. Cisco markets the first full-stack post-quantum network operating system, and Fortinet iterates hybrid IPsec across releases. Palo Alto sells cipher translation as a bridge for unupgradable applications, and carriers now offer quantum-resilient managed services. Purpose-built hardware has arrived, from quantum-optimized firewalls to routers with post-quantum interfaces.

The same boxes are also where the migration breaks. Middleboxes that assume a TLS hello fits in one packet silently drop the larger post-quantum handshakes, and Section 4 collects the measured failure modes. Wi-Fi, meanwhile, has barely started, with its standards group still working toward a first draft.

1. Next-Gen Firewalls

Firewalls terminate VPN tunnels, inspect encrypted traffic, and gate every packet into the enterprise, so their cryptography is load-bearing twice over. The major vendors have all shipped hybrid post-quantum key exchange for IPsec, and the race has moved on to inspection, management planes, and signatures. This section covers each vendor's position.

Recent Developments

The features are arriving monthly. Check Point added HTTPS inspection support for post-quantum traffic [1], shipped ML-KEM per the FIPS standard in a hotfix [2], and plans internal control-channel coverage by year end [3]. Fortinet extended post-quantum TLS, hybrid key exchange plus ML-DSA authentication, to the management interface itself [4]. Juniper's newest release signs software images with ML-DSA-87 per CNSA 2.0 across its major platform lines [5].

Cisco introduced a router generation purpose-built with quantum-safe processors and interfaces [6]. Palo Alto released quantum-optimized firewall hardware [7], added ML-KEM at all levels to its TLS profiles [8], and expanded certificate lifecycle and inventory features [9]. It also paired with AT&T on a quantum-resilient SASE fabric heading to general availability [10]. Even pfSense shipped post-quantum SSH key exchange [11].

Current State

Check Point runs hybrid IKEv2 in production. Its R82 release introduced Quantum-Safe Key Exchange, classical Diffie-Hellman plus ML-KEM through the standard IETF mechanism [12], carried into its appliance release notes [13]. Its community documentation is unusually candid about hardware: classical crypto-offload chips may not accelerate lattice operations, pushing post-quantum exchanges onto general CPUs [14]. It cites Germany's updated technical guideline as external validation [14], and cross-vendor interoperability works when both peers implement the same RFCs, with automatic classical fallback unless post-quantum is forced [15].

Fortinet has iterated longest. FortiOS 7.6 brought post-quantum IPsec with ML-KEM plus non-finalized candidates and QKD integration [16], implemented through the multiple-key-exchange RFCs with up to seven rounds [17]. Debug logs show real negotiations mixing ML-KEM with a fallback candidate in one session [18], and the feature has carried unchanged through four releases into 8.0 [19]. That release adds hybrid key exchange under deep inspection in both modes [20][21], covers the clientless VPN [21], and defaults post-quantum SSH on management [21].

Cisco is roadmap-first on the firewall line. ML-KEM for IPsec targets late 2026 releases [22], with decrypted inspection, QUIC, and remote-access VPN staged behind draft standards [22]. It assigns ML-DSA to authentication and image signing and holds SLH-DSA as deliberate algorithmic diversification [23], all under a portfolio-wide commitment through December 2026 [23].

Palo Alto sells the transition itself. Its Quantum-Safe Security solution combines discovery, risk assessment, and live cipher translation to ML-KEM [24]. PAN-OS 12.1 supports every NIST algorithm plus experimental candidates with agility features [25]. Its firewalls both translate quantum-safe sessions for classical internal applications and police post-quantum usage through decryption policy [26].

The open-source pair split by layer. pfSense is deliberately upstream-dependent, waiting on OpenSSL, FreeBSD, and strongSwan [27], while integrating validated wolfCrypt for the future [28]. OPNsense shipped something concrete: post-quantum SSH key exchange for administering the firewall itself [29]. Both inherit their IPsec future from strongSwan, whose stable plugin already carries ML-KEM, making the wait a packaging question [30].

Unresolved Conflicts

Juniper's naming needs care. One source groups its Quantum Buffer SSH feature with the FIPS post-quantum capabilities arriving in Junos [31]. Juniper's own documentation instead describes Quantum Buffer as extended classical parameters, a crypto-agility stopgap rather than a NIST algorithm [32]. The documentation is the better source, and a Juniper datasheet naming the algorithm would settle it.

2. ADCs & Load Balancers

Application delivery controllers terminate TLS for the applications behind them, which makes them the single upgrade that quantum-protects a whole server farm. All four major vendors now ship hybrid key exchange. This section covers the rollouts, and the first real interoperability bug of the migration.

Recent Developments

The configuration guides are the news. F5 published exact steps for enabling post-quantum cipher groups on its virtual servers [33]. A10 confirmed its hybrid support ships enabled by default on client-side TLS with no new hardware, through OpenSSL 3.5 [34].

Current State

F5's rollout came in stages. The pre-standard draft group arrived in late 2024, client-side only [35]. The finalized hybrid group landed in TMOS 17.5 for both sides [36], and the FIPS-curve variants followed in March 2026 alongside quantum-secure VPN tunneling [37]. Version 21.1 consolidated everything, including forward-proxy use [38].

Its implementation keeps classical certificates for authentication, costs roughly a quarter more CPU than classical exchange, and works with optional hardware security modules [39]. The migration's first notable interop bug is documented here: listing hybrid groups before classical ones broke TLS 1.2 handshakes with common test tools, with the fix being order [40].

Citrix claims the first-mover title. NetScaler previewed hybrid support in April 2025 and reached general availability that August, marketed as among the first production-ready platforms [41]. The feature binds the hybrid group to its enhanced SSL profile, framed around harvest-now-decrypt-later [42], and the platform sits on the U.S. defense procurement list [43].

Its marketing leans on survey data showing few organizations have any quantum roadmap [44]. A10 keeps it simple: hybrid key exchange in its operating system, classical and post-quantum in parallel, prioritizing key establishment over everything else [45][45].

Unresolved Conflicts

A10's default-on detail rests on a blocked source. The cipher groups and default enablement come from a social post whose retrieval failed verification, so they annotate rather than supersede the confirmed announcement [34][45]. A10's release notes would settle it.

3. Routers, Switches, SD-WAN

The routing and switching layer carries everything else, and its vendors are competing on how deep post-quantum goes: into boot chains, line-rate encryption, and the silicon itself. This section covers Cisco's full-stack push, the carrier services built on it, and the quieter moves at Nokia, Juniper, and Arista.

Recent Developments

Cisco declared full-stack and shipped pieces to match. IOS XE 26 is marketed as the first enterprise network OS with post-quantum across secure boot, IPsec, and management, secure by default [46]. ML-KEM-1024 IPsec ships on the WAN aggregation and edge router lines, with campus-switch parity a documentation check rather than a marketing claim [47]. A validated design extends the envelope across routing fabrics, combining post-quantum MACsec, IPsec, and ML-DSA boot anchors [48], and the next minor release adds image signing and switch-line coverage [49]. Context matters here: NIST's backup KEM and fourth signature remain in the pipeline [47].

The neighbors moved distinctively. Arista's documentation lists a pure post-quantum KEM category, non-hybrid, a framing unlike every other vendor covered, with no confirmed hybrid alternative alongside [50][50]. Juniper's Quantum Buffer feature rolls out across both its OS lines for phased adoption [51]. Nokia's ANYsec delivers line-rate quantum-safe MACsec extended over MPLS and segment routing [52], on specific silicon generations [53], with a partner supplying out-of-band key distribution [54].

Versa documented production post-quantum IPsec across its SASE platform from one codebase, verified by packet capture [55]. The AT&T-Palo Alto fabric bakes the hybrid RFCs into SD-WAN's data, management, and control planes [56].

Current State

Cisco's configuration documentation grounds the marketing. Post-quantum IKEv2 can enforce ML-KEM with no classical fallback, cannot touch legacy IKEv1, is unsupported on one VPN technology, and fragments at standard MTU with the largest parameter set [57]. Hybrid SSH combines ML-KEM with a classical curve, downgrade-proof by configuration [58]. Quantum-safe MACsec secures the LAN first hop, paired with WAN IPsec as defense in depth [59].

The published roadmap names the new hardware, routers, switches, and future access points, against dates through mid-2027 [60]. The smart switches anchor boot in ML-DSA-87 through a hardware trust module [61].

The others hold narrower positions. Juniper signs software images with ML-DSA-87 across six platform families [62] and has offered quantum-safe IPsec since 2024 via preshared keys and QKD interfaces [63]. It runs a key manager brokering static and quantum-sourced keys [64], and treats post-quantum math and QKD physics as parallel tracks [65]. Its strongest configuration aligns with CNSA 2.0 [66][67]. Nokia documents protecting utility-grid control traffic with hybrid IKEv2 as a deployable transitional design [68].

The market layer confirms demand. Industry analysts list post-quantum among the few features enterprise demand is actually driving in SD-WAN [69]. AT&T launched the first major North American carrier post-quantum SD-WAN service, built on Cisco's new routers and aimed at regulated sectors [70]. A specialist vendor positions a purpose-built quantum-safe alternative against retrofit approaches [71].

4. TLS Inspection & Handshake Size Breakage

Post-quantum handshakes are several times larger than classical ones, and a generation of network equipment was built assuming they would stay small. The result is the migration's most concrete operational pain: silent drops, extra round trips, and fragmentation. This section collects the measured failures and what actually mitigates them.

Recent Developments

The failure reports are now specific. One post-quantum handshake failed silently over a four-byte MTU mismatch with path discovery blocked [72]. Cellular paths impose stricter limits that fragment oversized handshakes into drops [73]. Distinct failure cliffs appear when handshakes cross roughly 10 and 30 kilobytes, where buffers give out [74].

Go's larger default hello has stalled middleboxes [75], and a full ML-DSA certificate chain can overflow the TCP initial congestion window, costing a round trip [76]. The IETF's application guidance formally warns about fragmented hellos and middlebox drops [77][73][73].

The inspection layer is adapting rather than breaking. Palo Alto firewalls read the hello's group list and can strip unsupported algorithms in forward proxy [78]. Fortinet inspects hybrid traffic in both modes [79], Cloudflare's gateway covers both legs [80], and F5 documents forcing classical fallback when sizes bite [81]. The measured base rate is reassuring: under one percent of real connections hit failures or extra round trips, mostly from inspection proxies and appliances with hardcoded limits [82].

Current State

The case studies define the genre. Meta's resumption hello outgrew a packet and broke fast-open assumptions, costing a full round trip, and the fix was downgrading to the smaller Kyber parameter to fit [83] [83]. Chrome's rollout measured a four percent median latency increase and exposed middleboxes hardcoding the one-packet assumption [84]. A catalog site names the three recurring patterns: rejected oversized hellos, truncating buffers, and parsers that fail without rejecting [85].

One corporate firewall dropped fragmented hellos silently for two days [86]. Even modern stacks have edges: a proxy's SNI routing failed outright on fragmented post-quantum hellos [87], and Check Point notes the CPU and packet growth plainly [88].

The structural answers are known. QUIC is less exposed because multi-packet handshakes are native and encrypted [86], though its anti-amplification rule delays certificate-heavy handshakes until address validation [89]. Measurement work finds the pain concentrates in tail latency under loss, fading as page sizes grow [90]. Practical guidance: measure handshake sizes against path MTU, test representative middleboxes, watch for silent retries, and roll out by traffic splitting, never a flag day [91].

Deep-inspection deployments feel it first, especially on tunneled links [92]. The policy tooling is arriving too, with Cato logging and enforcing post-quantum per rule [93] and Zscaler inspecting hybrid traffic inline [94].

5. Wi-Fi & WPA3

Wireless security rides on the same certificates and key exchanges as everything else, plus standards of its own. Wi-Fi is the least migrated major transport: the relevant IEEE group is still resolving comments toward a first draft. This section is short because the protocol work is young.

Recent Developments

The standards engine has started. IEEE task group P802.11bt is developing post-quantum authentication and key-management suites for Wi-Fi [95].

Current State

The study group's scoping tells the story. Formed in April 2025, it examines how the enterprise authentication methods Wi-Fi relies on could gain quantum-resistant KEMs through complementary IETF work [96]. WPA2 and WPA3 both rest on RSA and elliptic curves [96]. The surprising research result: a cross-layer analysis found WPA2-Personal in a stronger overall post-quantum posture than WPA3-Personal and WPA2-Enterprise, upending the assumption that newer means more quantum-resistant [97]. WPA3's 192-bit enterprise suite is a higher classical bar, not a post-quantum one [98].

Unresolved Conflicts

The draft status is one blocked source deep. A meeting report says the group resolved 22 comments and instructed its editor to begin Draft 1.0 [99], a more advanced status than the confirmed baseline [95]. Its retrieval failed verification. The group's published minutes would settle it.

6. VPN Concentrators & WAN Encryptors

At the network's long-haul edge sit the boxes that encrypt whole links: site-to-site VPN heads and the optical encryptors under oceans. This is where post-quantum meets terabit line rates, often layered with quantum key distribution. This section covers a category quietly merging into firewalls at one end and into light itself at the other.

Recent Developments

The optical trials went transatlantic. Colt and Ciena moved 800-gigabit quantum-safe traffic 6,900 kilometers across the Atlantic using NIST-compliant algorithms [100]. On the appliance side, FortiGate carries hybrid post-quantum IPsec with QKD options [101]. Its 7000G series is described as the first commercial VPN appliance implementing the FIPS 203 algorithms [102]. FortiOS 8.0 adds the preshared-key RFC as a second quantum-resistance layer down to the client [103].

Palo Alto supports ML-KEM site-to-site tunnels across current hardware generations with no replacement required [104]. Cisco's new router generation pushes quantum-safe IPsec throughput past 200 gigabits [6], and a startup sells a VPN built on one-time symmetric keys instead of PKI entirely [105].

Current State

The optical layer is shipping, not demonstrating. Ciena, Quantum Corridor, and Toshiba ran what they call the first 1.6-terabit quantum-safe encryption on a live commercial network, hybrid post-quantum plus QKD over production fiber [106]. No hardware swap was needed, with the new wavelength coexisting beside legacy encrypted traffic and line-rate AES in 3-nanometer silicon [107]. Ciena sells it as a standing capability with open QKD interfaces [108].

The history runs deep: ADVA has researched quantum-safe transport since 2015 with field trials since 2018, and markets its platforms as fully post-quantum equipped [109]. Colt's trial series spanned QKD, preshared keys, and post-quantum across European and transatlantic spans at extreme reliability [70]. Internet2 demonstrated FIPS-compliant protection across 1,390 backbone miles [110], and a European operator launched a quantum-safe connectivity service on Adtran optics [111]. PacketLight integrated the standards with hybrid options [112], and Ciena showed the full stack, PQC, QKD, and authentication, at 1.6 terabits on another platform [113].

The concentrator category is dissolving into adjacent boxes. Cisco frames VPN post-quantum as a firewall-platform feature with authentication following in 2027, advising buyers to weigh post-quantum secure boot now [22]. Cloudflare's WAN appliance went hybrid in February [114], and its IPsec service interoperates with Cisco and Fortinet gear for site-to-site tunnels [115]. AT&T's managed service wraps the Cisco hardware [116], and the ML-KEM foundation sits in the router OS beneath them all [117].

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